MULTIMODAL DEEPFAKE DETECTION STUDIO

Detailed Project Report

Architecture, model inventory, evidence, evaluation, deployment readiness, and roadmap

Report scope	Source-grounded review of the checked-in Multimodal-deepfake repository
Prepared	22 October 2026
Primary system	Xception + Wav2Vec 2.0 + audiovisual SyncStream with multi-task late fusion
Audio modernization	Legacy Wav2Vec baseline; AASIST and AASIST-L verified; SSL-AASIST and XLSR-MamBo adapters staged for GPU environments
Evidence basis	Source code, configurations, JSON/CSV evaluation artifacts, model cards, audit documents, and existing result figures

Purpose

This report consolidates the project as it is implemented. It explicitly separates (1) measurements stored in this repository, (2) published/upstream reference values reproduced in project documentation, and (3) integrations whose interface is coded but whose checkpoint or runtime is unavailable locally. This distinction is essential for a forensic system: high benchmark scores must not be interpreted as an unconditional guarantee on new media.

Decision summary: retain the multimodal fusion model for video-level analysis, retain the legacy Wav2Vec checkpoint as a reproducible baseline, and use AASIST/AASIST-L as the currently verified standalone audio upgrade. Run SSL-AASIST only in its prescribed Colab/Linux GPU environment and calibrate it on a representative development set before production use.

Classification: research and decision-support system; not an autonomous forensic determination service.

1. Executive summary

The repository implements two complementary detection paths. The multimodal path analyses video, speech, and audiovisual synchrony, emitting three task-specific logits: visual manipulation, audio manipulation, and any manipulation. The standalone path decodes an audio recording to 16 kHz mono, slices it into overlapping windows, applies voice-activity gating, scores windows with a selected detector, and merges suspicious intervals for review.

Area	Current state	Evidence / interpretation
Multimodal detector	Implemented; historical evaluation artifact present	Three-stream late fusion. Stored FakeAVCeleb test artifact reports AUC 0.9994 for the overall head across 3,276 held-out videos. The checkpoint itself is not currently present in models/.
Existing Wav2Vec baseline	Implemented and checkpoint verified	Actual legacy checkpoint uses 512-dim CNN acoustic features, a 512 -> 256 -> 2 MLP, 2.0 s windows, 1.0 s hop, and a dev-calibrated spoof threshold of 0.984982.
AASIST / AASIST-L	Integrated, checkpointed, parity-tested	Raw-waveform SincNet + graph-attention alternatives. AASIST-L trades some upstream benchmark accuracy for lower CPU latency.
SSL-AASIST	Adapter and Colab training path implemented; local execution blocked	Requires Fairseq-compatible Linux/CUDA runtime, XLSR-53 frontend and upstream parity validation. No local measured score is claimed.
XLSR-MamBo	Optional adapter scaffold only	Dedicated CUDA 12.8 / FlashAttention / Mamba environment and a compatible checkpoint are required; unsupported on this Windows CPU environment.

Key qualification

The repository contains both a legacy checkpoint contract and a newer generic Wav2Vec adapter contract. They are not interchangeable. The verified legacy wav2vec_best.pt expects 512-dimensional CNN features and two logits, whereas src/deepfake_detector/model.py defines a newer 768-dimensional transformer-hidden-state AudioClassifier with one logit. Any deployment must choose the detector/configuration pair deliberately rather than mixing these interfaces.

2. End-to-end architecture

The multimodal model deliberately separates independently manipulable signals. This lets the verdict identify whether a face/video stream, speech/audio stream, or cross-modal synchrony is responsible rather than treating every clip as a single opaque binary class.

Input media	Preprocessing	Feature streams	Decision
Video: MP4/MOV/AVI Audio: native track or standalone file	8 sampled frames; MTCNN face crop (299 x 299); lip crop (112 x 112); 16 kHz mono; 64-band log-mel	Image: Xception + temporal mean/std pooling Audio: frozen Wav2Vec 2.0 + projection Sync: 3D lip CNN + 2D mel CNN	3 x 512-d concatenation -> MLP trunk -> video / audio / any heads

Multimodal neural components

Component	Implementation	Function
ImageStream	timm Xception; frame features are pooled by mean and standard deviation, then projected to 512-d	Detects spatial/temporal facial manipulation evidence.
AudioStream	Frozen facebook/wav2vec2-base-960h; 768-d mean pooled transformer state; 768 -> 512 -> 512 projection	Carries learned speech/acoustic evidence into fused video analysis.
SyncStream	3D CNN on 8 lip crops + 2D CNN on log-mel; 128 -> 512 projection	Tests audiovisual consistency, including lip-motion/speech mismatch signals.
Fusion / heads	1536 -> 512 -> 256 MLP with dropout; 3 binary output heads	Returns distinct logits for visual fake, audio fake, and overall manipulation.

Experimental fusion extension

training/cross_attention.py implements a v4 CrossAttentionDetector. It loads/freeze-warm-starts image, audio, and sync streams from v3, then applies three four-head cross-attention blocks (each modality attends to the other two), residual LayerNorm, a new fusion trunk, and the same three heads. It is an implemented training experiment; no checked-in v4 checkpoint or result artifact establishes a deployed performance claim.

3. Model inventory and status

This inventory includes the requested existing Wav2Vec model and all newly implemented model paths visible in the detector registry and training suite. 'Verified' means code/checkpoint/evidence exists in the repository; it does not mean the model is universally calibrated for arbitrary media.

Model	Architecture / input	Status	Deployment position
Legacy Wav2Vec 2.0 + MLP	Frozen Wav2Vec CNN extractor; mean-pool 512-d features; MLP 512 -> 256 -> 2; 2.0 s windows	Verified checkpoint and native/sliding-window artifacts	Baseline and compatibility reference
Multimodal late fusion	Xception + Wav2Vec 2.0 transformer + 3D/2D SyncStream; three heads	Implemented; test/ablation artifacts present; current weight file absent from models/	Video and audiovisual analysis
AASIST	Raw 16 kHz waveform; 70-filter Sinc frontend; spectro-temporal graph attention + master node; 4.04 s windows	Integrated, official checkpoint available, upstream parity reported	Active verified standalone audio upgrade
AASIST-L	Compact AASIST raw-waveform graph-attention variant; 4.04 s windows	Integrated, official lightweight checkpoint available	Lower-latency standalone alternative
SSL-AASIST	XLSR-53 (Wav2Vec 2.0 multilingual SSL) + weighted layer aggregation + AASIST backend; RawBoost training	Registry adapter, config and Colab training script implemented; no compatible local runtime/checkpoint measurement	Primary GPU fine-tuning candidate
XLSR-MamBo	XLSR SSL frontend + bidirectional Mamba state-space backend	Optional adapter implemented; CUDA-specific dependencies missing	Research challenger only
Cross-attention v4	Cross-modal attention over frozen image/audio/sync embeddings	Training implementation only; no checked-in trained output	Multimodal research experiment

3.1 Existing Wav2Vec baseline: exact operational contract

The legacy wav2vec_best.pt is the reproducible baseline, not merely a conceptual Wav2Vec model. Audio is resampled to 16 kHz mono and framed into 32,000 samples (2.0 seconds) with a 16,000-sample hop. Windows below RMS 0.015 are gated as non-speech. The backbone feature extractor is mean-pooled to a 512-dimensional vector, classified by Linear(512,256), BatchNorm, ReLU, Dropout(0.3), Linear(256,2), and scored as softmax class index 1 (spoof). The calibrated operating threshold is 0.9849822521, selected on ASVspoof 2019 LA development data.

> The newer generic AudioClassifier source is materially different: it uses mean-pooled 768-dimensional transformer hidden states, LayerNorm, GELU layers, and one logit. It must be paired only with an aligned audio_model.pt-type checkpoint, not with wav2vec_best.pt.

3.2 Newly integrated graph-attention models

AASIST and AASIST-L receive raw 16 kHz waveform windows of exactly 64,600 samples (4.0375 seconds), repeat-pad shorter recordings, advance at a 32,000-sample (2.0 second) hop, and use a two-logit output convention: spoof index 0 and bonafide index 1. Their native score orientation is higher_is_bonafide, so a fake verdict arises below the bonafide threshold. The service intentionally does not invent a fake probability from uncalibrated logits.

4. Standalone audio detection service

The detector registry exposes legacy_wav2vec_mlp, aasist, aasist_l, ssl_aasist, and xlsr_mambo behind an AudioDetector interface. This isolates model-specific score conventions and input contracts from the user-facing pipeline.

Step	Operational behavior	Why it matters
1. Decode	soundfile first; imageio-ffmpeg and librosa fallbacks; convert to 16 kHz mono normalized waveform	Supports common uploaded formats without requiring torchaudio.
2. Window	Generate overlapping fixed-length windows using detector requirements; zero padding for legacy, repeat padding for AASIST-family	Preserves the original model's input contract.
3. Gate	RMS VAD normally excludes low-energy windows; falls back to all windows when whole recording is quiet	Avoids silence dominating recording-level evidence.
4. Score	Batched detector inference returns raw scores, orientation, per-window fake score, and optional bonafide score	Prevents class-index/score-direction errors between models.
5. Aggregate	Mean recording score; top suspicious windows; merge consecutive flagged intervals	Produces an analyst-readable timeline rather than a single unexplained label.
6. Report	Verdict, threshold, duration, windows, intervals, warnings, model identity/hash prefix, timing	Enables traceability and UI/CLI/JSON consumption.

Score semantics are a safety-critical boundary

Legacy Wav2Vec produces a spoof probability where higher means more fake. AASIST, AASIST-L, SSL-AASIST, and XLSR-MamBo expose bonafide-oriented logits where lower means more fake. The service reads score_orientation on every prediction and reverses the decision rule accordingly. This design is tested to protect against a common but severe integration error: treating a bonafide score as a fake probability.

User-facing integration: Flask provides status plus upload endpoints for multimodal video and standalone audio; scripts/analyze_video.py provides command-line analysis and JSON export. If the relevant checkpoint is missing, the UI is designed to report availability rather than silently return a fabricated result.

5. Data governance and evaluation protocol

The audio evaluation protocol uses a nine-column manifest schema: sample_id, path, label, dataset, split, speaker_id, source_id, attack_type, and language. The canonical label convention is 0 = genuine and 1 = fake. The project distinguishes native benchmark inference from end-to-end sliding-window recording inference, preventing application-processing artifacts from being confused with raw model discrimination.

Dataset evidence	Count / split	Notes
ASVspoof 2019 LA	121,461 total: 25,380 train; 24,844 dev; 71,237 eval	12,483 genuine / 108,978 fake in stored summary; eval has unseen A07-A19 attacks.
Local evaluation	6 recordings: 4 genuine, 2 fake	Small, in-the-wild compressed-media sanity set only; unsuitable for firm population metrics.
Leakage audit	PASS	Stored audit reports no cross-split speaker, source, or exact-duplicate overlap in the audited manifests.
Multimodal FakeAVCeleb test	3,276 videos / 75 unique identities	Reported in README/result artifact as subject-disjoint held-out evaluation.

Protocol controls

Threshold selection is limited to the development split and is then applied to evaluation data without label peeking. The leakage checker also records model training provenance. These practices preserve a meaningful separation between fitting, calibration, and evaluation. The data are heavily spoof-skewed (approximately 89.72% fake in the stored summary), so ROC AUC, PR AUC, EER, genuine false-positive rate, and per-attack recall should be read alongside accuracy.

6. Measured evaluation results

Numbers in this section are reproduced from checked-in JSON/CSV artifacts. They are not fresh reruns performed for this report. Published reference values are called out separately.

6.1 Multimodal historical test artifact (FakeAVCeleb)

Head	Accuracy	Balanced accuracy	ROC AUC	F1	Recall
Video	0.9988	0.9994	0.9991	0.9994	0.9987
Audio	0.9869	0.9869	0.9982	0.9875	0.9860
Any / overall	0.9969	0.9919	0.9994	0.9984	0.9972

The artifact is associated with epoch 3. The overall head performs strongly across categories, but RealVideo-FakeAudio is the weakest stored category for overall accuracy (0.9333, n=75); this is consistent with audio-only manipulation being harder when visual evidence remains genuine. Interpret the very high scores in the context of the dataset and split, not as a claim of universal robustness.

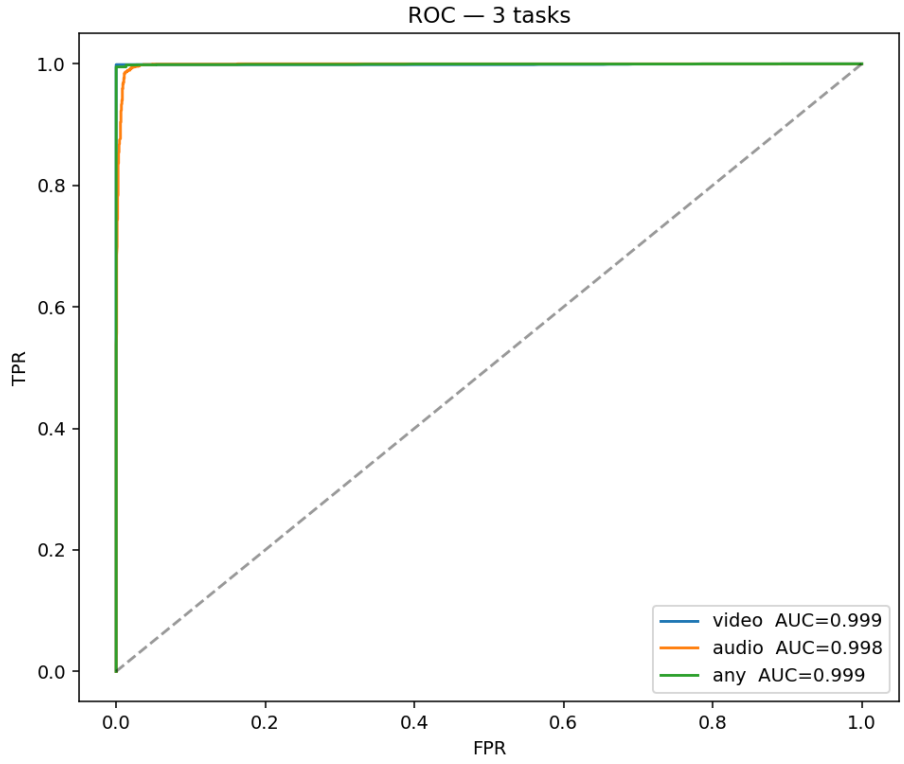


Figure 1. Stored ROC curves for video, audio, and overall tasks.

6.2 Multimodal ablation artifact

Configuration	Overall AUC	Overall F1	Interpretation
Full multimodal	0.9994	0.9984	Best stored complete configuration.
No image	0.7618	0.7070	Largest measured degradation; visual stream carries strong signal in this dataset.
No audio	0.9993	0.9987	Near-parity on this data; does not establish audio is unnecessary for audio-only attacks outside the split.
No sync	0.9931	0.9877	Synchrony stream adds measurable support.
Only audio	0.6393	0.4493	Audio alone is insufficient for the predominant video-fake test composition.

7. Audio model results and interpretation

The legacy Wav2Vec evaluation used its native 512-dimensional feature benchmark path and a threshold selected on the ASVspoof development partition. It is therefore the clearest comparable local baseline. Its 7.13% EER on the held-out ASVspoof 2019 LA evaluation split is paired with high AUC (0.9815) but meaningful weak-attack failure modes.

Metric (ASVspoof 2019 LA eval, n=71,237)	Legacy Wav2Vec + MLP
EER	7.13%
Accuracy at dev-calibrated threshold	84.90%
ROC AUC / PR AUC	0.9815 / 0.9978
Spoof recall / genuine FPR	83.33% / 1.47%
Confusion matrix	TN 7,247; FP 108; FN 10,647; TP 53,235

Legacy per-attack diagnostic

Unseen attack	Spoof recall	Attack EER	Reading
A07-A14	96.8%-99.9% (except none below 96.8%)	0.37%-2.07%	Strong on several vocoder/TTS/VC variants.
A15	88.0%	4.55%	Performance begins to decline.
A16	88.2%	5.29%	Waveform-concatenation vulnerability.
A17	27.1%	18.93%	Major failure mode; direct concatenation.
A18	38.1%	13.42%	Major voice-conversion failure mode.
A19	52.8%	12.67%	Major voice-conversion failure mode.

AASIST comparison: evidence levels

Measure	Legacy	AASIST	AASIST-L
ASVspoof 2019 LA eval EER	7.13% (stored local artifact)	0.83% (published/upstream reference)	0.99% (published/upstream reference)
Local 6-recording ROC AUC	0.5000	0.7500	0.7500
Local 6-recording EER	75.00%	50.00%	50.00%
Local CPU latency / recording	1,368 ms	6,278 ms	3,112 ms

The local set has only six recordings, so its AUC/EER values are diagnostic—not deployable benchmark estimates. It does, however, reveal a real domain-shift issue: the legacy score ranking collapses on compressed web audio, while AASIST-family scores provide better ranking separation but have a downward bonafide-logit shift at the standard 0.0 operating threshold. A calibrated development set from the target media domain is required before hard decisions are trusted.

8. Runtime, checkpoints, and deployment readiness

Asset	Evidence	Readiness
wav2vec_best.pt	666,067 bytes; SHA-256 efba2b1fadcb602b928b49496d6bd0ce9786fc10f8d28f76a6de77dfd378159f	Verified legacy baseline checkpoint under initial setup/deepfake_phase1/checkpoints/.
AASIST.pth	1,281,532 bytes; SHA-256 51d2d9cf0738172f61e2a384ec50a54a55363240f67c971ed55a92435bc1a1c0	Verified standalone upgraded backend.
AASIST-L.pth	426,428 bytes; SHA-256 814331d088032bb4c3fa61cc014789eadeed464209dd094ab3a2dd6ffbdce27a	Verified low-latency standalone backend.
models/best_model.pt and audio_model.pt	Documented as expected paths but models/ currently holds only README.md	Multimodal and generic standalone Wav2Vec service needs checkpoint placement/configuration before live deployment.
SSL-AASIST / XLSR-MamBo	No local compatible runtime/checkpoint assertion	Use isolated GPU environment; do not expose as a ready production choice locally.

Environment constraints

The audited local environment is Windows 11, Python 3.13, CPU-only PyTorch, with no CUDA device. Legacy Wav2Vec and AASIST-family CPU inference are feasible, though AASIST is slower. SSL-AASIST requires the upstream Fairseq/XLSR lineage and is blocked locally by Python/Fairseq packaging and C++ extension constraints; its designated execution target is Linux Google Colab with CUDA and at least 12 GB VRAM. XLSR-MamBo additionally needs CUDA 12.8, FlashAttention, Mamba-SSM, causal-conv1d, and Hydra.

Production recommendation

For current deployment, wire checkpoint paths explicitly and surface model status in the UI. Use AASIST-L where CPU latency matters and AASIST where its full model is acceptable. Keep the legacy Wav2Vec result available for baseline comparison, but do not treat it as robust to all voice-conversion/concatenation attacks. Escalate uncertain or high-impact cases to analyst review with the interval report and multimodal evidence.

9. Risks, limitations, and prioritized roadmap

Risk / limitation	Observed implication	Mitigation
Distribution shift	Compressed social/web audio shifts AASIST logit distributions; local set is tiny.	Collect representative dev audio; calibrate thresholds; evaluate by codec, language, device, and attack family.
Legacy blind spots	Low recall on A17-A19 attack types.	Adopt AASIST as current alternative; execute SSL-AASIST/RawBoost evaluation on GPU.
Missing checkpoints	UI can be unavailable for multimodal/generic audio paths.	Version and hash approved weights; configure MODEL_PATH and AUDIO_MODEL_PATH; add startup verification.
Score comparability	Softmax probability and raw bonafide logits are not directly comparable.	Preserve orientation metadata; calibrate each backend independently; never average incompatible scores.
Forensic misuse	False positives/negatives can affect people and decisions.	Use as decision support; retain provenance; provide uncertainty and human review.

Recommended next sequence

1. Make the existing checkpoint inventory deployable: configure absolute/approved paths, perform a startup load test, and retain checkpoint hashes in every report. 2. Create a larger representative local-development set separated into calibration and final test partitions. 3. Calibrate AASIST/AASIST-L operating points on development data, including codec-aware validation. 4. Run the supplied SSL-AASIST Colab workflow with upstream Fairseq parity checks, RawBoost training, and held-out ASVspoof 2021 / in-the-wild evaluation. 5. Benchmark latency/throughput and uncertainty by audio length. 6. Train/evaluate v4 cross-attention only after the baseline multimodal checkpoint and

dataset split are reproducibly packaged.

10. Conclusion

The project is a well-structured multimodal deepfake-detection research system with a usable separation between media preprocessing, model backends, scoring semantics, reporting, and UI/CLI integration. Its strongest present engineering achievement is the evidence-aware modernization of standalone audio detection: the legacy Wav2Vec model is preserved with its true contract, AASIST and AASIST-L are integrated with parity and provenance evidence, and advanced SSL-based candidates are staged without being misrepresented as locally complete. The highest-value remaining work is deployment packaging, domain-specific calibration, and GPU evaluation of the SSL-AASIST path.

Appendix: primary repository evidence consulted

Area	Key files
Core implementation	src/deepfake_detector/model.py; audio_detection/service.py; audio_detection/registry.py; detector adapters
Configuration	configs/audio/legacy.yaml; aasist.yaml; ssl_aasist.yaml
Measured artifacts	results/test_results.json; results/ablation.csv; reports/legacy_eval_report.json; sliding-window JSON reports
Governance/audits	docs/current_system_audit.md; docs/evaluation_protocol.md; docs/model_card.md; reports/leakage_audit.json
Training extensions	training/cross_attention.py; training/train_ssl_aasist_colab.py; notebook_v2/colab_ssl_aasist_quickstart.md